

Member ID: _____

Time: _____

Rank: _____



FUNDAMENTALS OF WEB DESIGN (415)

STATE 2023

Production:

Multiple Choice (20 @ 5 points each) _____ (100 points)

Application _____ (480 points)

TOTAL POINTS _____ **(580 points)**

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Directions: Identify the letter of the choice that *best* completes the statement or answers the question.

1. CSS stands for?
 - A. Cascading Style Sheet
 - B. Color Style Sheet
 - C. Cascade Style Sheet
 - D. Color Sheet Styles
2. H1 is called a ____ in CSS
 - A. attribute
 - B. selector
 - C. element
 - D. tag
3. The ____ property is used in CSS to define all four sides of an elements border in a single declaration.
 - A. border-width
 - B. padding
 - C. border
 - D. collapse
4. The ____ property is used to set the width of an element's bottom border.
 - A. border-width
 - B. width
 - C. border-bottom-width
 - D. down width
5. The ____ CSS property is used to set the color of an element's four side border.
 - A. color
 - B. border-color
 - C. bg color
 - D. border all
6. The ____ CSS property is used to set the text formatting.
 - A. text-decoration
 - B. font
 - C. style
 - D. font-style

7. The property used for the dotted border around an image is ____.
 - A. border-color
 - B. border-decoration
 - C. border-style
 - D. border-line
8. The ____ property used to display border around a cell without any content.
 - A. empty-cell
 - B. blank-cell
 - C. non-content-cell
 - D. void-cell
9. If the width of a table adjusts to the current width of the browser window, then ____ is the width of the table.
 - A. 100%
 - B. 1024 px
 - C. 540 px
 - D. 180 px
10. ____ is used to make all paragraph elements 'RED' in color
 - A. p.all{color:red;}
 - B. p.all{color:#990000;}
 - C. all.p{color:#998877;}
 - D. p{color:red;}
11. ____ is the default value of the Position attribute
 - A. fixed
 - B. absolute
 - C. inherit
 - D. relative
12. The ____ CSS property is used to control the text size of an element
 - A. font-style
 - B. text-size
 - C. text-style
 - D. font-size
13. To add margin between the DIV border and its inner text _____ CSS property is used.
 - A. spacing
 - B. padding
 - C. margin
 - D. inner margin

14. Which element allows the list items to appear with discs, circles or square types before the item?
- A. ol
 - B. ul
 - C. dt
 - D. dd
15. The four main tags that are included in table elements include:
- A. table, th, tr, td
 - B. table, ta, tb, tc
 - C. table, th1, th2, th3
 - D. table, th, ti, tl
16. What is the syntax to create a hyperlink?
- A. link
 - B. src
 - C. href
 - D. hyper
17. What interactive state describes how a hyperlink appears after the user has previously clicked on the link.
- A. link
 - B. hover
 - C. active
 - D. visited
18. What interactive state describes how a hyperlink appears when the user is actively clicking on the link.
- A. link
 - B. hover
 - C. active
 - D. visited
19. The syntax to use email in a hyperlink is: text
- A. link
 - B. email
 - C. mailto
 - D. post

20. The ____ tags are used to group content elements as blocks.

- A. <dividers></dividers>
- B. <div></div>
- C. <wrap></wrap>
- D. <wrapper></wrapper>

APPLICATION

During this exercise, you are only allowed to use **Notepad**. The use of an online editor or any use of the Internet will result in immediate disqualification.

Screen capture of the expected end result is included for your inspection and is to be used as a guide for completing this exercise.

When complete, submit your files generated for this exercise (HTML and Style Sheet). Also, you will need to submit a screenshot (using the snipping tool or screenshot tool) of your finished work as it appears in the web browser so that it may be used for grading purposes.

Your task is to regenerate the web page shown in the screen capture provided in this packet, using HTML5, CSS, and the parameters specified below.

1. Overall Specifications

- a. The documents must be valid HTML5.
- b. The documents must be styled with an external cascading style sheet.
 - 1. ALL styling MUST be done with the style sheet **UNLESS SPECIFICALLY NOTED.**
- c. **The document must not use tables for any sort of layout or positioning.**

2. File Creation

- a. Create a folder using your contestant number as the name of the folder.
- b. Ensure you have the “wheat.jpg” from the start files in your project folder.
- c. Create two new files called index.html and a stylesheet named styles.css; these files should be saved to your project folder

3. General Page Specifications (doctype, HTML, and head)

- a. The opening HTML element must include the language attribute with the value for the English language; there should also be an ending html tag.
- b. Title should have **Your Contestant Number – Fundamentals of Web Design.**
- c. The head tag must have a meta element with the character set an attribute with a value of UTF-8.
- d. The head tag must have a meta element with viewport as the name and the appropriate content attributes.
- e. You should link your style sheet along with defining the relationship and type.
- f. Lastly, ensure you have your opening and closing body tag.

4. Banner Image

- a. The first opening and closing div will contain an img and another opening and closing div
- b. The container div should contain an id called “container”
- c. Inside the container div, the first item should be a “wheat.jpg” img with the alt text of “up close picture of wheat” along with an inline style setting the width to 100%
- d. Next, a new div will contain the text: “Welcome to Kansas!” and contain an id called “centered”

5. Whitespace

- a. The next element will be an opening and closing div
- b. There should be nothing inside the two div tags but a “whitespace” class should be given to the opening div

6. Intro Paragraphs

- a. The next section will be a div containing an h1, and two paragraphs
- b. The opening div should contain the “text” class
- c. Inside the div should be an h1 tag containing the text: “Intro”
- d. Next, will be two paragraphs.
 1. The first paragraph should contain:
 - o Kansas, situated on the American Great Plains, became the 34th state on January 29, 1861. Its path to statehood was long and bloody: After the Kansas-Nebraska Act of 1854 opened the two territories to settlement and allowed the new settlers to determine whether the states would be admitted to the union as “free” or “slave,” North and South competed to send the most settlers into the region. This quickly led to violence, and the territory became known as “Bleeding Kansas.”
 2. The second paragraph should contain:
 - o Kansas has long been known as part of America’s agricultural heartland and is home to the major U.S. military installation Fort Leavenworth. In 1954, it became a battleground for civil rights when the landmark Brown v. Board of Education of Topeka case was decided in the Supreme Court, ending the doctrine of “separate but equal” in public schools. Kansas is also known for its contributions to jazz music, barbecue and as the setting of L. Frank Baum’s classic children’s book The Wizard of Oz.

7. Whitespace

- The next element will be an opening and closing div
- There should be nothing inside the two div tags, but a “whitespace” class should be given to the opening div

8. State Info

- The next section will contain state info in a table
- The first element should be an opening div with an id of “middle” ;
- Next, an h1 tag should contain the text: “State Info”
- After the h1, a table tag should have nine table rows with two table data cells in each row
- Below is a table of what each row and cell should contain:

Date of Statehood:	January 29, 1861
Capital:	Topeka
Population:	2,853,118 (2010)
Size:	82,278 square miles
Nicknames:	Sunflower State; Wheat State; Jayhawk State
Motto:	Ad astra per aspera (“To the stars through difficulties”)*
Tree:	Cottonwood
Flower:	Wild Native Sunflower
Bird:	Western Meadowlark

*Please note that the Motto row will have italics around the following underlined text: To the stars through difficulties ; the quotes and parentheses should NOT be italicized

9. Whitespace

- The next element will be an opening and closing div
- There should be nothing inside the two div tags, but a “whitespace” class should be given to the opening div

10. Interesting Facts

- The Interesting Facts section will contain a div enclosing a h1 heading, an unordered list with six list items. It will also contain a br followed by a paragraph allowing users to click to return to the top along with a final br
- The first element should be opening div with a class called “text”
- Next, an unordered list should be started and contain the following list items:
 - Fort Riley was established near the Kansas River in 1853 to protect settlers and trade along the Oregon and Santa Fe Trails. In 1866, the 7th Cavalry organized at the fort under General George Armstrong Custer, who later led the regiment in the infamous attack on the Sioux and Cheyenne tribes at the Battle of Little Bighorn in June of 1876.
 - When French astronomer Pierre Janssen first discovered the element, he called helium on the sun in 1868, it was believed to be one of the rarest elements. It wasn’t until 1905, when faculty members at The University of

Kansas in Lawrence began experimenting with gas from a newly drilled well in Dexter, that helium was identified as a common element found on Earth.

- Kansas is the leading producer of wheat in the United States. Referred to as “The Wheat Capital of the World,” Sumner County produced 9 million bushels in 2009.
- Meade’s Ranch in Osborne County, Kansas, is the Geodetic Center of North America—the point of reference by which all property lines and boundaries in North America are surveyed. Identified in 1901, this triangulation station for the United States, Canada and Mexico is also known as the North American Datum.
- Although the origin remains unclear, the terms “jayhawk” and “jayhawker” were originally used to describe raiders and looters during the period of unrest following the Kansas-Nebraska Act of 1854 when settlers of the Kansas Territory were left to decide the issue of slavery. It was later used solely as a label for free-state proponents, and eventually became associated with all Kansans.
- During the Civil War, Kansas suffered the highest rate of fatal casualties of any Union state.

d. Add a break tag after the closing unordered list tag

e. Next, add a paragraph tag with an id of “return”

1. The paragraph should contain the following text:

- o “- Return to top by clicking here -“
- o The word ‘here’ should have an anchor tag surrounding it with a href to the id “container”

11. CSS

a. body selector should contain:

1. margin set to 0 px

b. container id should contain:

1. position set to relative
2. text alignment should be set to center
3. text color should be white

c. centered id should contain:

1. position set to relative
2. top set to 50%
3. left set to 50%
4. transform element given a translate(-50%, -50%) value
5. font size set to 50 px
6. font family given Verdana, then Geneva, then Tahoma with a backup of sans-serif
7. text shadow with a h-shadow and v-shadow of 3px along with a blur of 5px and color of black

d. h1 selector should contain:

1. text alignment should be set to center

2. font family given Verdana, then Geneva, then Tahoma with a backup of sans-serif
- e. text class should contain:
 1. width set to 80%
 2. margin set to auto
 3. text alignment should be set to justify
- f. whitespace class should contain:
 1. width set to 100%
 2. height set to 200px
- g. table selector should contain:
 1. border with a width of 1px, solid style and black color
 2. border spacing is 0px
 3. margin left is auto
 4. margin right is auto
- h. td selector should contain:
 1. padding set to 10px
 2. border with a width of 1px, solid style and black color
- i. tr selector with a pseudo class nth-child(odd) should contain:
 1. background color of #CACACA
- j. tr selector with a pseudo class nth-child(even) should contain:
 1. background color of #FFF
- k. td selector with a pseudo class first-child should contain:
 1. font weight bold
- l. middle id should contain:
 1. background color of #f7efab
 2. top padding of 50 px
 3. bottom padding of 100px
- m. return id should contain:
 1. text alignment should be set to center
 2. font style set to italic
- n. Two media queries should be added:

```
@media only screen and (min-width: 601px) {  
    div#centered {  
        font-size: 30px;  
    }  
}
```

```
@media only screen and (max-width: 600px) {  
    div#centered {  
        font-size: 80px;  
    }  
}
```

SAMPLE



Intro

Kansas, situated on the American Great Plains, became the 34th state on January 29, 1861. Its path to statehood was long and bloody: After the Kansas-Nebraska Act of 1854 opened the two territories to settlement and allowed the new settlers to determine whether the states would be admitted to the union as "free" or "slave," North and South competed to send the most settlers into the region. This quickly led to violence, and the territory became known as "Bleeding Kansas."

Kansas has long been known as part of America's agricultural heartland and is home to the major U.S. military installation Fort Leavenworth. In 1954, it became a battleground of the civil rights movement when the landmark *Brown v. Board of Education of Topeka* case was decided in the Supreme Court, ending the doctrine of "separate but equal" in public schools. Kansas is also known for its contributions to jazz music, barbecue and as the setting of L. Frank Baum's classic children's book *The Wizard of Oz*.

State Info

Date of Statehood:	January 29, 1861
Capital:	Topeka
Population:	2,853,118 (2010)
Size:	82,278 square miles
Nicknames:	Sunflower State; Wheat State, Jayhawk State
Motto:	<i>Ad astra per aspera ("To the stars through difficulties")</i>
Tree:	Cottonwood
Flower:	Wild Native Sunflower
Bird:	Western Meadowlark

Interesting Facts

- Fort Riley was established near the Kansas River in 1853 to protect settlers and trade along the Oregon and Santa Fe Trails. In 1866, the 7th Cavalry organized at the fort under General George Armstrong Custer, who later led the regiment in the infamous attack on Sioux and Cheyenne tribes at the Battle of Little Bighorn in June of 1876.
- When French astronomer Pierre Janssen first discovered the element he called helium on the sun in 1868, it was believed to be one of the rarest elements. It wasn't until 1905, when faculty members at The University of Kansas in Lawrence began experimenting with gas from a newly drilled well in Dexter, that helium was identified as a common element found on Earth.
- Kansas is the leading producer of wheat in the United States. Referred to as "The Wheat Capital of the World," Sumner County produced 9 million bushels in 2009.
- Meade's Ranch in Osborne County, Kansas, is the Geodetic Center of North America — the point of reference by which all property lines and boundaries in North America are surveyed. Identified in 1901, this triangulation station for the United States, Canada and Mexico is also known as the North American Datum.
- Although the origin remains unclear, the terms "jayhawk" and "jayhawker" were originally used to describe raiders and looters during the period of unrest following the Kansas-Nebraska Act of 1854 when settlers of the Kansas Territory were left to decide the issue of slavery. It was later used solely as a label for free-state proponents, and eventually became associated with all Kansans.
- During the Civil War, Kansas suffered the highest rate of fatal casualties of any Union state.

- Return to top by clicking [here](#) -

APPLICATION RUBRIC

HTML	Point Value	Score
Doctype declaration is correct	5	
<html></html> is in correct location with en designation	5	
<head></head> is in correct location	5	
<title></title> is in correct location and contestant # is present	5	
<meta> contains charset for UTF-8	5	
<meta> contains name (viewport) and content (width...)	5	
<link> tag reference to a CSS style sheet	15	
<body></body> is in correct location	5	
First <div> contains an id called container; the first div encloses a tag and another opening and closing <div>	15	
The uses “wheat.jpg”, has alt text “up close picture of wheat” and an internal width style set to 100%	5	
The opening <div> (inside the container div) contains an id called “centered” and between the opening and closing div contains text “Welcome to Kansas!”	5	
Second structural <div> contains the whitespace class with no content inside the opening and closing div	5	
Third structural <div> contains the text class and encloses an <h1> tag, and two <p> tags	15	
The <h1> tag contains the text “Intro”	5	
Paragraphs in Intro section are typed correctly with no mistakes (-1 point for each mistake for max of -10) and are located in the correct location	15	
Fourth structural <div> contains the whitespace class with no content inside the opening and closing div	5	
Fifth structural <div> contains an id called “middle” and encloses an <h1> tag, 9 <tr> tags that each have two <td> tags	15	
The <h1> tag contains the text “State Info”	5	
Each <tr> contains two <td> tags. Content in <td> is type correctly with no mistakes. Each incorrect <tr>, <td>, <td> grouping is -5 points.	45	
Sixth structural <div> contains the whitespace class with no content inside the opening and closing div	5	
Seventh (last) structural <div> contains the class “text” and encloses an <h1> tag, tag with six tags, along with a then <p> tag then finally another tag	15	
The <h1> tag contains the text “Interesting Facts”	15	
The tag is present and contains 6 tags	5	
Each tag is typed correctly with no mistakes (-1 point for each mistake for max of -20) and are located in the correct location	30	
 tag after closing tag	5	

The <p> tag contains the text “- Return to top by clicking here -” with the word “here” having an <a> tag. The anchor tag links to the container id	15	
 tag after the closing <p> tag	5	
HTML TOTAL	280	

CSS	Point Value	Score
Style Sheet is external and named styles.css	5	
- body selector contains: - margin set to 0 px	5	
- container id contains: - position set to relative - text alignment should be set to center - text color should be white	15	
- centered id contains: - position set to relative - top set to 50% - left set to 50% - transform element given a translate(-50%, -50%) value - font size set to 50 px - font family given Verdana, then Geneva, then Tahoma with a backup of sans-serif - text shadow with a h-shadow and v-shadow of 3px along with a blur of 5px and color of black	35	
- h1 selector contains: - text alignment should be set to center - font family given Verdana, then Geneva, then Tahoma with a backup of sans-serif	10	
- text class contains: - width set to 80% - margin set to auto - text alignment should be set to justify	15	
- whitespace class contains: - width set to 100% - height set to 200px	10	
- table selector contains: - border with a width of 1px, solid style and black color - border spacing is 0px - margin left is auto - margin right is auto	20	
- td selector contains: - padding set to 10px - border with a width of 1px, solid style and black color	10	
tr selector with a pseudo class nth-child(odd) contains:	10	

o background color of #CACACA		
• tr selector with a pseudo class nth-child(even) contains: o background color of #FFF	10	
• td selector with a pseudo class first-child contains: o font weight bold	10	
• middle id contains: o background color of #f7efab o top padding of 50 px o bottom padding of 100px	15	
• return id contains: o text alignment should be set to center o font style set to italic	10	
@media only screen and (min-width: 601px) { div#centered { font-size: 30px; } }	10	
@media only screen and (max-width: 600px) { div#centered { font-size: 80px; } }	10	
CSS TOTAL	200	

Categories	Point Value	
HTML	280	
CSS	200	
TOTAL	480	